
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Despite the remarkable overall fidelity of the images they produce, generative image models trained on large-scale datasets systematically fail to render seemingly elementary concepts—e.g., human hands or objects appearing in groups of four—that are abundantly and faithfully represented in their training data. To date, such failures have been catalogued only anecdotally, leaving open the central question of whether they are isolated idiosyncrasies or the surface manifestations of a structural limitation common to models trained on internet-scale corpora. We settle this question by introducing a systematic, fully unsupervised methodology that identifies and characterizes “conceptual blindspots”—concepts present in the training distribution yet absent from, or distorted in, a model’s generations. Our method leverages sparse autoencoders (SAEs) to extract interpretable concept embeddings and, through them, to quantify and directly compare the prevalence of each individual concept between real and generated images. As the representational backbone of this comparison, we train an archetypal SAE (RA-SAE) on DINOv2 features spanning 32,000 concepts—the largest such SAE reported to date—which affords the concept-level resolution needed to localize precisely where a model’s output distribution diverges from the natural image distribution. Applied to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), our approach pinpoints concrete suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees), and it further reveals that suppression is the dominant and systematically shared failure mode across every model we evaluate. Descending to the level of individual datapoints, the same framework isolates memorization artifacts—instances in which models reproduce highly specific visual templates encountered during training. Together, these results establish a theoretically grounded and model-agnostic framework that diagnoses generative image models by measuring the conceptual fidelity of their outputs against the underlying data-generating process.

1 Introduction

Generative image models trained on large-scale datasets have attained unprecedented capabilities, driving applications both within the vision domain [1–9] and well beyond it [9–15]. This success notwithstanding, a growing body of qualitative [16–18] and quantitative [19–20] evidence establishes that these models routinely fail to render even elementary concepts—among them human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. Tellingly, when prompted for such concepts, these models do not omit them outright but instead synthesize closely related yet incorrect structure—rendering, for



Figure 1: $\delta(k)$ quantifies a model’s tendency to over- or under-generate a concept c_k relative to its frequency in the natural data. Concepts with $\delta(k) < 0.1$ are designated **suppressed conceptual blindspots** and concepts with $\delta(k) > 0.9$ **exaggerated conceptual blindspots**. The images shown, produced by four widely used generative image models, juxtapose concepts falling into each blindspot regime against concepts the same models render faithfully, making the two failure regimes directly legible at a glance.

instance, hands with six fingers. We term these failure modes “conceptual blindspots”¹ and their existence is genuinely counterintuitive: the training corpora contain abundant, faithful demonstrations of the very concepts that the models miss. This tension motivates the central question of our work: are such failures the idiosyncratic quirks of a handful of concepts, or the signature of a systematic phenomenon in which, across a broad spectrum of concepts, models are consistently either unable or overly inclined to generate them? We show that the latter holds.

Existing approaches for evaluating generative image models are structurally ill-suited to answering this question [28, 29]. The prevailing metrics are coarse-grained scores of image realism—chief among them FID [30]—which aggregate over entire images and are therefore constitutively blind to the concept-level distributional failures at issue. Diversity- and coverage-oriented metrics such as CLIPScore [31–33] move closer, yet they too operate at the granularity of whole images or prompts rather than that of individual features or concepts [34, 35], and consequently cannot localize which concepts a model fails to reproduce. Qualitative protocols—participant surveys [5, 17, 36–38] and open-ended exploration [39]—can surface individual conceptual failures, but they do not constitute a scalable, repeatable methodology that transfers across model families and supports systematic, quantitative comparison.

This work. We close this gap by developing an automated, unsupervised methodology that identifies concepts in the data distribution whose odds of being generated by the trained model diverge from their odds under the true data-generating process. Crucially, rather than merely quantifying aggregate distribution coverage, the methodology pinpoints which concepts constitute the mismatch and resolves, for each, whether it is suppressed or exaggerated. Our contributions are three-fold:

- **A formalization of conceptual blindspots in generative image models.** We cast conceptual blindspots as measurable discrepancies between the concept content of natural images and that of a model’s generations, and we derive from first principles a quantitative measure of these discrepancies grounded in the data-generating process (Section 3). This formalization replaces anecdotal and human-defined concept evaluations with a single, calibration-free criterion, which we prove both concentrates rapidly from modest sample sizes and preserves the ordering of the underlying energy gap.
- **A scalable, unsupervised method for detecting conceptual blindspots with sparse autoencoders.** We operationalize this formalization as a concrete, model-agnostic pipeline that uses sparse autoencoders (SAEs) to extract and directly compare concept distributions between natural and synthesized images (Section 4). To this end, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features spanning 32,000 concepts—the largest such RA-SAE reported to date—which supplies the concept-level resolution the analysis demands and yields seed-independent, reproducible concept bases.
- **An exploratory tool and a multi-level empirical analysis.** We release an interactive web tool that supports distribution-, datapoint-, and composition-level analysis of blindspots across models (Sections 5.1–5.4), and we apply our method to four widely used models—Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5). The analysis surfaces concrete suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents)

¹We borrow the term “blindspots” from the psychology literature [27], in which it denotes scenarios where an agent makes biased decisions despite exposure to observations that contradict the rationale behind those decisions.

77 and exaggerated conceptual blindspots (e.g., wood background texture and palm trees),
78 illustrated in Fig. 1 and traces them to shared training-data properties and model-specific inductive
79 biases alike.

80 The model, code, and web tool are available at a software repository.

81 2 Related Work

82 2.1 Explainability in Vision

83 Early work in explainable AI, computer vision included, centered on attributing predictions to influ-
84 ential input regions [40, 42]. Such attribution methods, however, convey little semantic information
85 about the learned representations and can yield explanations that are demonstrably unfaithful [43, 45].
86 Concept-based interpretability [46] emerged to overcome these shortcomings by identifying semanti-
87 cally meaningful directions in a network’s representation space, thereby revealing not merely where a
88 model attends but which concepts and structures it encodes [47–49].

89 Recent work establishes that the leading concept-based interpretability methods—ACE [44],
90 CRAFT [48], and sparse autoencoders (SAEs) [50, 51]—all instantiate a single underlying dictionary-
91 learning problem under differing constraints [52]. Among them, SAEs are the most scalable route
92 to concept-based interpretability. Although the original formulation exhibits well-documented
93 limitations—overly specific features [53], compositionality challenges [54], and limited interven-
94 tion effects [55]—subsequent refinements address these directly, most notably the archetypal SAE
95 (RA-SAE) [56] that we adopt and scale here.

96 Complementary interpretability techniques—prompt-based probing [57], attention-map and activa-
97 tion visualizations [58, 59], and dataset-level statistics such as diversity or distribution-coverage
98 metrics [32, 33]—each afford only partial purchase on our question. Their shared limitation is one of
99 granularity: they operate over whole images or prompts rather than over the fine-grained features
100 and concepts that blindspot analysis requires [34, 35]. They further rely on subjective interpretation
101 and do not disambiguate distinct failure modes [60]. In short, existing methods and metrics are
102 inadequate for systematically localizing the feature- and concept-level weaknesses of generative
103 image models [28]—precisely the gap our framework is designed to close.

104 2.2 Generative Image Models

105 Diffusion-based methods now dominate generative vision modeling across modalities, spanning
106 image [4, 61, 64], video [65–72], and 3D [7, 73–75] synthesis. In image generation, the lineage
107 traces to denoising diffusion probabilistic models (DDPMs) [76], subsequently generalized to non-
108 Markovian diffusion processes by denoising diffusion implicit models (DDIMs) [77].

109 The Stable Diffusion (SD) [9] family brought DDPMs to the research and open-source communities at
110 scale and was extended through successive releases—SD 2 [78], SD 3 [79], and SD XL [80]. A large
111 ecosystem of architectural modifications followed, imposing additional constraints on the diffusion
112 process—over style [81, 82], pose [83], and identity [84, 85], among others—and admitting alternative
113 input modalities such as image-to-image generation. A parallel line of latent diffusion models built
114 atop SD—Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and FLUX [89]—has proliferated in turn,
115 and it is precisely this architectural diversity that our model-agnostic methodology is designed to
116 compare on a common footing.

117 2.3 Datasets for Generative Image Models

118 The recent success of generative vision models rests largely on the confluence of abundant compu-
119 tation and large-scale internet datasets [90–92]. LAION-5B [93] has been pivotal to the training
120 of open-source text-to-image models, SD and its derivatives among them. Scraped from Common
121 Crawl [94], it comprises over 5B image-caption pairs, 2.3B of them in English; other prominent cor-
122 pora include COYO-700M [95] and Conceptual Captions [96], with 700M and 3M pairs, respectively.

123 The dataset’s prominence has been accompanied by mounting concerns over its biases [97–101].
124 Filtering efforts notwithstanding, harmful content persisted [97, 99]—including NSFW material [100]
125 and hundreds of CSAM instances [101]—prompting the corpus’s temporary withdrawal from official

channels. The data is further marked by low-quality images [102] and internet-style captions (e.g., product descriptions) that diverge sharply from how users actually prompt the trained models [103]. These distributional properties of the training corpus are, we argue, a direct cause of the conceptual blindspots our analysis is designed to expose, and our framework makes their downstream effects measurable.

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning seeks sparse representations in which each sample is reconstructed as a linear combination of a few dictionary atoms [104–106]. Grounded in compressed sensing theory [107–108], the field has progressed from early vector quantization [109] to a rich family of formulations, including non-negative matrix factorization [110–111], sparse PCA [112], and K-SVD [113]. More recent developments encompass online algorithms [114], structured sparsity [115], and theoretical recovery guarantees [116–117], together with increasingly tight connections to deep learning [118–119].

Sparse coding has, in turn, been brought to bear on the emergence of high-level concepts inside diffusion models [120]. Prior to diffusion models, concept-grounded interpretability was applied to earlier generative architectures through concept-bottleneck models, which demand human intervention at training time [121], and post-hoc detectors that retrofit concept supervision [122]. Both paradigms, however, presuppose a fixed, human-defined concept vocabulary and therefore cannot register trends the user has not anticipated in advance—the very trends that an unsupervised method is uniquely positioned to surface.

For the evaluation of generative image models specifically, aggregate metrics—precision, recall, density, and coverage [35, 123]—and latent density scores that predict sample quality from the model’s latent space [124] have been proposed. These reliably expose distributional gaps in aggregate, yet they remain silent on which specific concepts are under- or over-represented.

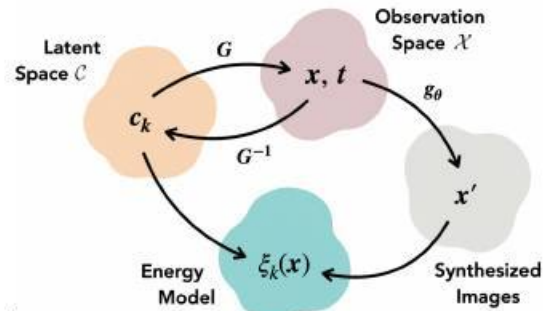
Taken together, these limitations motivate a scalable, unsupervised framework that systematically identifies and quantifies concept-level failure modes in generative image models [125]—the framework we develop next.

3 Formalizing Conceptual Blindspots in Generative Models

We begin by formalizing *conceptual blindspots* as systematic discrepancies between the conceptual content of natural images and that of a model’s generations. This formulation yields a principled, quantitative measure that identifies precisely which concepts a generative model under- or over-represents relative to its data distribution; the overall process is illustrated in Fig. 2. While the derivation invokes standard assumptions from the identifiability and disentanglement literature [126–130], we find empirically that the framework elicits meaningful phenomenology even when those assumptions are only approximately satisfied, establishing that its diagnostic value is robust to their exact satisfaction rather than contingent on it.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G: \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational economy, we let $p(\cdot)$ denote both the latent density $p(c)$ and its pushforward to image space $p(x)$, where $x = G(c)$. The invertibility of G is what legitimizes this overloading: it induces, through the change-of-variables formula, a well-defined distribution over $\mathcal{X}$. The individual dimensions of the latent space thus constitute the **Concepts** underlying the data distribution $P_{\mathcal{X}}$, defined over the observation space of images $\mathcal{X}$. Concretely, distinct latents may encode concepts such as color, shape, size, and location [131, 132]. We additionally let the data-generating process associate a text description $t \in \mathcal{T}$ with each sampled image, although we do not model this association explicitly. These descriptions serve as the condi-



4

Figure 2: **Schematic of our framework.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations (x, t) by the invertible DGP. Conditioned on the caption t , S_θ synthesizes an image x' . The energy model

tioning signal for a text-conditioned **Generative image model** g_θ with parameters θ : trained on a corpus of image-text pairs, g_θ maps a noise signal $\eta \sim \mathcal{N}(0, I)$ together with a scene description t to an image x that realizes that description.

Defining conceptual blindspots in g_θ therefore reduces to a single, well-posed comparison: the probability mass a concept receives under the data-generating process versus the mass it receives under the model. To operationalize this comparison, we take an evaluation set of natural image-text pairs $(\mathcal{D}_\mathcal{X}, \mathcal{D}_\mathcal{T})$ and construct a matched set of synthesized images $\mathcal{D}'_\mathcal{X}$ from the identical text descriptions. We estimate a concept’s probability of occurrence through an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps an image to a d -dimensional vector whose k^{th} coordinate is the scalar energy of the k^{th} concept, i.e., $\xi_k(x) = E(c_k)$. It is precisely the invertibility of the data-generating process that renders these estimates recoverable. The probability the process assigns to an image x then follows as $p_k(x) \propto \exp(-\sum_k \xi_k(x))$ (with Z_ξ the partition function), which aggregates to the population-level estimate $p_k(\mathcal{D}_\mathcal{X}) = \prod_{x \in \mathcal{D}_\mathcal{X}} p_k(x)$. Composing this with the sigmoid map $\sigma(\cdot)$, we now define the average energy difference in the k^{th} concept between the datasets $\mathcal{D}_\mathcal{X}$ and $\mathcal{D}'_\mathcal{X}$.

Definition 2 (Energy Difference). Let $x \in \mathcal{D}_\mathcal{X}$ denote a real image sampled from the data-generating process G , and let $x' \in \mathcal{D}'_\mathcal{X}$ be a synthetic image generated by the model g_θ . Let $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the energy difference for concept k as:

$$\begin{aligned} \delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{x'}[\xi_k(x')] - \mathbb{E}_x[\xi_k(x)]) \\ &= \frac{p_k(\mathcal{D}'_\mathcal{X})}{p_k(\mathcal{D}_\mathcal{X}) + p_k(\mathcal{D}'_\mathcal{X})}, \end{aligned} \quad (1)$$

where the expectations are taken over $\mathcal{D}'_\mathcal{X}$ and $\mathcal{D}_\mathcal{X}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{x \in \mathcal{D}} \xi_k(x))$ denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

The energy difference in the k^{th} concept thus expresses the relative probability that the concept occurs in a target set of observations (here, $\mathcal{D}'_\mathcal{X}$) versus a baseline set (here, $\mathcal{D}_\mathcal{X}$), normalized to $(0, 1)$ by the sigmoid so that 0.5 denotes exact parity. This single scalar furnishes a direct, interpretable criterion for conceptual blindspots.

Definition 3 (Suppressed / Exaggerated Conceptual Blindspots). Given a generative image model g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

In sum, a conceptual blindspot is a concept whose likelihood of occurrence in generated images departs markedly—through either suppression or exaggeration—from its prevalence under the data-generating process. Suppressed concepts activate disproportionately rarely ($\delta(k) < \lambda_{\min}$), whereas exaggerated concepts are systematically overrepresented ($\delta(k) > \lambda_{\max}$); throughout, we set $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to demarcate these two regimes.

This definition generalizes and sharpens the classical notion of “mode collapse” studied in past work (e.g., Bau et al. [39]), from which it differs chiefly in granularity. Whereas mode collapse concerns the suppressed or exaggerated odds of generating *entire images*, our measure isolates the shifted odds of *individual concepts*. A model’s failure to produce objects on a `white background`, for instance, registers in our formalism as a suppressed conceptual blindspot localized to that specific concept, rather than being obscured within a global, image-level collapse.

4 Method: Operationalizing the Definition of Conceptual Blindspots

We now instantiate the formalization of Sec. 3 as a concrete, reproducible pipeline for identifying conceptual blindspots in a generative model g_θ . The construction requires exactly two ingredients: (i) a set of images sampled from g_θ that can be placed in one-to-one correspondence with samples from

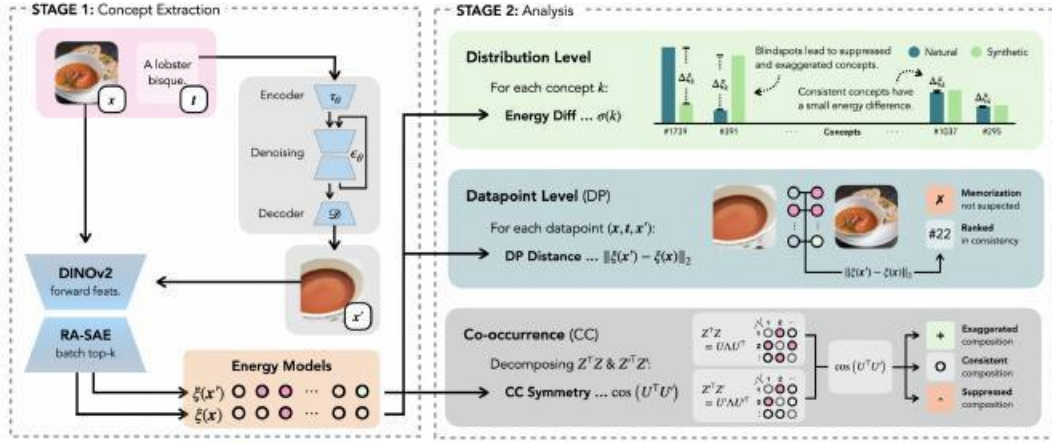


Figure 3: **Concept extraction pipeline.** For a triplet (x, t, x') , concepts are extracted in two stages: we first compute the DINOv2 features of each image and then map them through the RA-SAE into sparse concept embeddings, which constitute the energy models $\xi(x)$ and $\xi(x')$. By construction, the k^{th} coordinate $\xi_k(x) = E(c_k)$ holds the energy of the k^{th} concept, so that real and generated images become directly comparable within a shared concept basis.

For notation, $\|\cdot\|_F$ denotes the Frobenius norm and $\|\cdot\|_0$ the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix X , $X \geq 0$ denotes element-wise non-negativity. For $n > 0$, we write $[n] := \{1, \dots, n\}$ and denote the i -th row of a matrix A by A_i .

From Prompts to Latent Representations. To identify conceptual blindspots in g_θ , we compare a dataset $\mathcal{D}_X$ of image-caption pairs (x, t) drawn from the data-generating process G against their synthetic counterparts, generated by g_θ from the identical text descriptions, so that each real image is placed in exact correspondence with a prompt-aligned generation.

Concretely, given a caption t , we synthesize the counterpart image x' with a pretrained text-to-image generator $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$, realized as a denoising diffusion probabilistic model (DDPM) [78, 86, 87]. Sampling proceeds in latent space along the reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The terminal latent γ_0 is decoded by a pretrained VAE to produce the synthetic image $x' = \text{VAE}(\gamma_0)$. For the remainder of the paper we treat g_θ as a black box that maps prompts to images, $t \mapsto x'$; this deliberate abstraction is precisely what renders the method architecture-agnostic and applicable, without modification, across every diffusion model we study.

Defining the Energy Model. Prior work establishes that self-supervised learning methods can invert the data-generating process and recover its underlying energy function up to a linear transformation [126, 128, 133, 134]; we exploit this property directly by adopting DINOv2 [135], a state-of-the-art self-supervised encoder, as the backbone of our energy model. Because the number of concepts underlying the DGP is expected to exceed the dimensionality of the encoder’s feature space [51, 136], these concepts are necessarily represented in superposition; we therefore train sparse autoencoders (SAEs) on the DINOv2 features to disentangle the subspaces corresponding to individual concepts [50, 56, 137–139]. The rationale is exact: if DINOv2 encodes the generative concepts along approximately orthogonal directions—consistent with the independence structure posited in Def. 1—then an SAE trained on its features isolates each concept along a distinct latent dimension [136]. The activation the SAE assigns to that dimension then serves as our estimator of the ground-truth energy of the concept it encodes.

Formally, let $f : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the feature extractor (the DINOv2 model); we compute features $a = f(x) \in \mathbb{R}^d$ for both the natural and the synthetic images in $\mathcal{D}_X$ and $\mathcal{D}'_X$. Writing n for the number of image-text pairs, we stack the real and generated features into matrices $A, A' \in \mathbb{R}^{n \times d}$. Each feature vector is then decomposed into a sparse combination of learned concept atoms by an SAE. Let $D \in \mathbb{R}^{d \times K'}$ be a dictionary of K' concept vectors and $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ the SAE encoder that maps features to sparse codes. Applying Ψ row-wise produces the activation matrix

262 $\mathbf{Z} = \Psi(\mathbf{A}) \in \mathbb{R}^{n \times K'}$, whose i -th row $\mathbf{z}_i = \Psi(\mathbf{a}_i)$ is the concept decomposition of the i -th image.
 263 The decoder reconstructs features through $\mathbf{Z}\mathbf{D}^\top$, and the SAE is trained to minimize reconstruction
 264 error subject to sparsity and non-negativity constraints:

$$\min_{\Psi, \mathbf{D}} \|\mathbf{A} - \Psi(\mathbf{A})\mathbf{D}^\top\|_F^2 \quad \text{s.t.} \quad \Psi(\mathbf{A}) \geq 0, \|\Psi(\mathbf{A})_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

265 Vanilla SAEs are prone to converging on arbitrarily oriented dictionaries, which renders downstream
 266 analyses acutely sensitive to the random seed. To eliminate this instability and thereby guarantee
 267 seed-independent, reproducible results, we employ the *Archetypal* SAE (RA-SAE) [56] under a TOP-
 268 K sparsity constraint [137]. RA-SAE constrains each dictionary atom to be a convex combination of
 269 the training data. Formally, we write $\mathbf{D} = \mathbf{W}\mathbf{A}$ with $\mathbf{W} \in \Omega_{K', n}$, the set of row-stochastic matrices
 270 in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \left\{ \mathbf{W} \in \mathbb{R}^{K' \times n} \mid \mathbf{W} \geq 0, \mathbf{W}\mathbf{1} = \mathbf{1} \right\}. \quad (3)$$

271 Consequently, every atom $\mathbf{D}_i$ lies within the convex hull of the data $\text{conv}(\mathbf{A})$, and any reconstruction
 272 $\mathbf{Z}\mathbf{D}^\top$ resides within the conic hull $\text{cone}(\mathbf{A})$. This constraint tethers the learned concepts to the
 273 support of the data distribution and thereby guarantees that they remain faithful to it [56]. Once
 274 trained, the SAE yields a single, consistent concept basis in which both real and generated images are
 275 jointly expressed: $\mathbf{Z}$ for the real images and $\mathbf{Z}'$ for their generated counterparts. *Because both sets of*
 276 *codes are expressed over the same interpretable concepts, they render prompt-conditioned visual*
 277 *semantics directly comparable, and each concept’s activation supplies precisely the energy value on*
 278 *which our blindspot analysis operates.* Our method therefore constitutes a structured pipeline that,
 279 from a prompt and its associated real image $(\mathbf{t}, \mathbf{x})$, returns two sparse concept vectors $(\mathbf{z}, \mathbf{z}')$ and
 280 thereby places the real and generated visual content within a common conceptual basis for direct,
 281 concept-level comparison.

282 To summarize the full pipeline: beginning from a captioned image $(\mathbf{t}, \mathbf{x})$, we synthesize a counterpart
 283 $\mathbf{x}'$ and embed both images into a shared sparse concept space through the vision encoder and
 284 the trained SAE, obtaining interpretable representations $(\mathbf{z}, \mathbf{z}')$ that constitute the foundation for
 285 quantifying the conceptual shifts induced by the generative process.

286 5 Results

287 We analyze four widely used generative image models trained on LAION-5B—SD 1.5,
 288 SD 2.1, PixArt, and Kandinsky—over $|\mathcal{X}| = 10,000$ image-text pairs and their
 289 prompt-matched generations (Fig. 4). The analysis proceeds at three complementary
 290 and mutually reinforcing levels (Fig. 3): a ● **distribution-level** evaluation reveals
 291 suppressed and exaggerated concepts; a ● **datapoint-level** analysis surfaces failures at-
 292 tributable to ambiguity, omission, and memo-
 293 rization; and a ● **compositional** analysis ex-
 294 poses structured distortions in the geometry of
 295 concept co-occurrence.

297 At the center of our contribution is an interactive
 298 exploratory tool, shown in Fig. 5 a web-based
 299 interface organized around a UMAP projection of
 300 concept representations that supports the visu-
 301 alization and direct comparison of concept-level
 302 energy differences. It is publicly available at a
 303 software repository, together with pre-computed energy-difference data for all four evaluated models
 304 (SD 1.5, SD 2.1, PixArt, and Kandinsky), and every analysis reported below is derived through it. Its
 305 principal capabilities are:

- 306 • **Contrasting models and architectures.** For each evaluated model, the tool renders a UMAP
 307 visualization spanning all 32,000 RA-SAE concepts, with every scatter point representing
 308 an individual concept color-coded by its energy difference, so that entire blindspot profiles
 309 can be compared across architectures at a glance.



Figure 4: **Representative image-caption pairs** from LAION-5B, each shown alongside the images synthesized from the identical prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

- **Inspecting concepts.** Each concept is documented by a card that reports its key statistics, its most representative real and generated images (x , x'), and its co-occurrence patterns.
- **Exploring blindspots.** Beyond the UMAP and per-concept views, the tool provides global rankings of suppressed and exaggerated blindspots, foregrounding the most pronounced conceptual blindspots for immediate inspection.

Rather than exhaust any single phenomenon, we report a set of high-level findings that together demonstrate the breadth of the tool and open the door to broader, user-directed exploration.

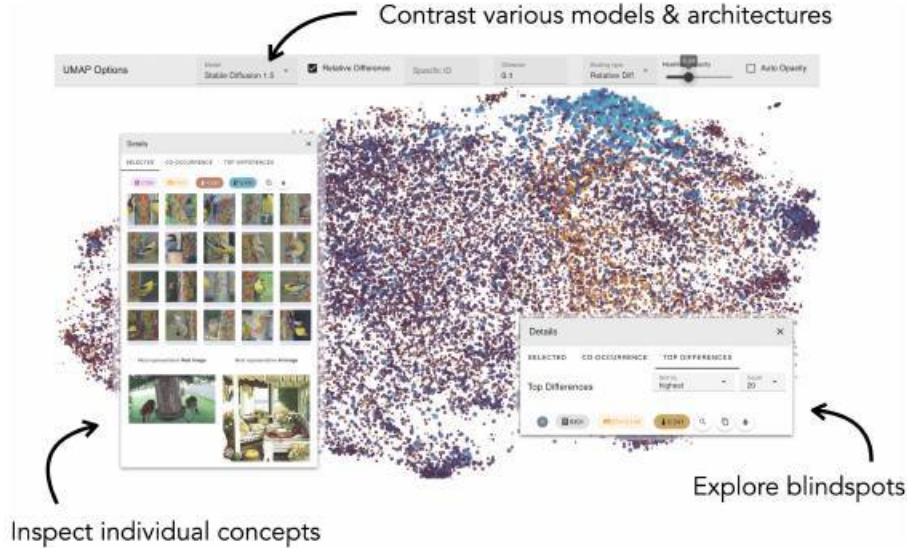


Figure 5: **Overview of the exploratory tool.** The web interface displays a per-model UMAP projection in which each dot is a concept color-coded by its energy difference. Selecting a concept opens a detail panel that reports its statistics and its most representative natural and generated images, x and x' , together with an ordered list of the concept’s co-occurrences and the global blindspot rankings, so that distribution- and datapoint-level evidence can be interrogated within a single interface.

5.1 Generative Image Models Cannot Produce Unseen Concepts

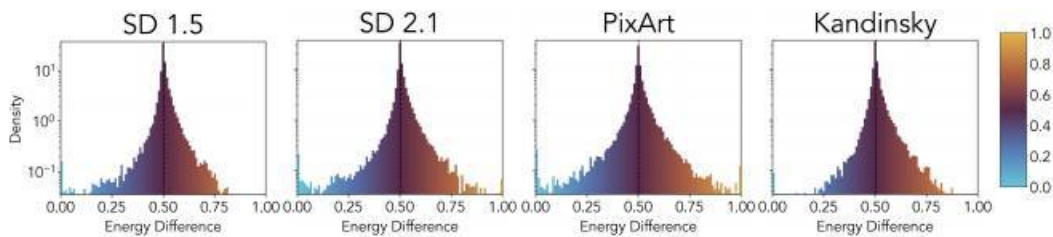


Figure 6: **Concept energy distribution.** Log-scale histograms of the energy difference $\delta(k)$ over all 32,000 concepts, contrasting the natural and synthesized distributions for each evaluated model. Mass to the left corresponds to suppressed (under-represented) concepts and mass to the right to exaggerated (over-represented) concepts; the pronounced left skew common to all four histograms establishes a systematic, cross-model bias toward suppression.

To characterize the disparities between the generative models and the natural image distribution, we first analyze the marginal energy difference $\delta(k)$ across all 32,000 concepts learned by the RA-SAE. As defined in Sec. 4, this quantity measures the relative prevalence of each concept in the synthesized versus the natural image set: $\delta(k) < 0.1$ marks concept k as under-represented (suppressed) in the generated images, and $\delta(k) > 0.9$ marks it as over-represented (exaggerated). Fig. 6 presents the distribution of $\delta(k)$ for each of the four models. Every model exhibits a heavy-tailed histogram with substantial mass at both extremes, establishing that concept-coverage discrepancies are systematic

rather than incidental. Decisively, the left tail—the suppressed concepts—is consistently denser and longer than the right across all four models, evidencing a robust and shared tendency toward concept suppression. This asymmetry is quantified directly by the uniformly negative skewness of the four distributions: -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

While all four models thus display both suppressed and exaggerated concepts, their detailed profiles diverge in informative ways. PixArt, for instance, exhibits the widest spread, consistent with a more heavily suppressed concept distribution. The left skew common to every distribution nonetheless points to a shared propensity for concept omission that transcends any single architecture; the finer structure of this behavior is the subject of the sections that follow.

5.2 Structure and Specificity of Conceptual Blindspots

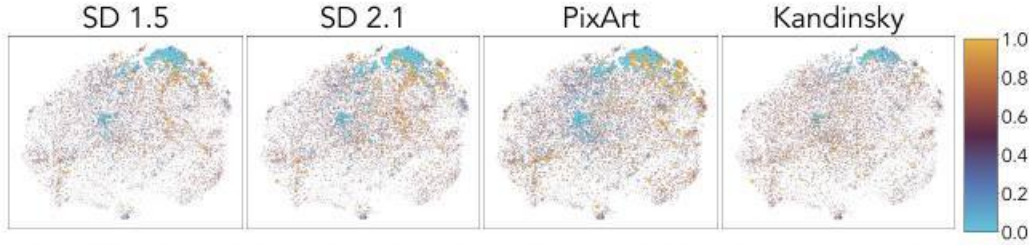


Figure 7: **Structure of concept energy differences.** UMAP visualizations of the 32,000 concepts, colored by their energy difference $\delta(k)$ between the natural and synthesized distributions. The emergence of coherent clusters—rather than scattered points—reveals that conceptual blindspots are spatially organized in concept space, with suppressed concepts concentrated at the blue end and exaggerated ones at the yellow end.

Whereas the previous section quantified the marginal discrepancies in concept frequency, we now characterize their global structure by embedding all 32,000 concepts into two dimensions via UMAP on the sparse codes, again coloring each concept by its $\delta(\cdot)$ value. As Fig. 7 shows, distinct concept clusters emerge for every model. These clusters frequently coincide with contiguous regions of conceptual blindspots—most conspicuously for suppressed (blue) concepts—demonstrating that blindspots are not randomly distributed but spatially structured, and hence traceable to shared biases in the training distribution or to architectural priors. To determine how consistent these blindspot patterns are across models, we analyze both the magnitude and the structure of the concept-level $\delta(\cdot)$ values. Fig. 8 reports scatter plots and pairwise Pearson correlations between the $\delta(k)$ vector of SD 1.5 and those of the remaining models. SD 1.5 and SD 2.1 are strongly correlated ($r = 0.82$), exactly as their shared architecture and training pipeline would predict. Their correlations with PixArt and Kandinsky, by contrast, are markedly lower ($r = 0.41$ and $r = 0.46$, respectively), establishing that these architectures allocate their blindspots to substantially different regions of the conceptual space.

Taken together, this analysis establishes that blindspots occupy two distinct regimes: some are universally shared—most plausibly a consequence of dataset properties—whereas others are strongly

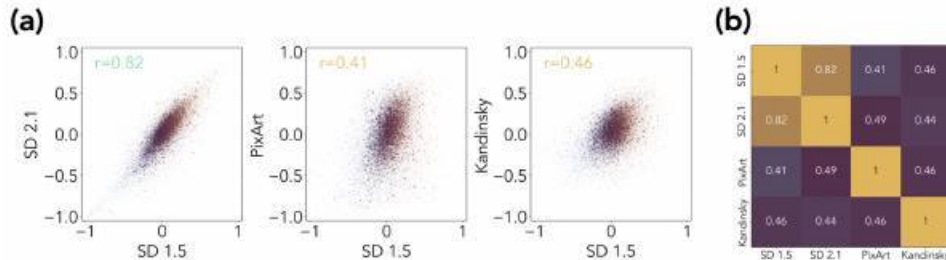


Figure 8: **Cross-model concept energy correlation.** Pairwise scatter plots of $\delta(k)$ across the four evaluated models, with Pearson correlation coefficients reported at top left. The strong alignment between SD 1.5 and SD 2.1 ($r = 0.82$) stands in sharp contrast to the substantially weaker correlations with the other architectures ($r = 0.41$ and $r = 0.46$), quantifying the extent to which blindspots are model-specific. **Correlation matrix of conceptual blindspots.** Heatmap of the pairwise Pearson correlations of δ across all models, summarizing how much of each model’s blindspot profile is shared with the others.

model-specific, arising from idiosyncrasies of training dynamics or model capacity. This dichotomy directly motivates the separate treatment, in the sections that follow, of blindspots shared across models and those unique to individual models.

5.3 Qualitative Examples of Blindspots

We now examine concrete instances of both suppressed and exaggerated blindspots in order to characterize the concepts that populate each regime. Fig. 9a presents a blindspot suppressed by every model: all four evaluated models fail to reproduce the concept **solid white on documents**. Although the caption references this concept explicitly, none of the generations recover the intended visual semantics—direct evidence that this region of concept space is systematically under-sampled across the entire set of models. Fig. 9b, by contrast, isolates a model-specific blindspot: the concept **pan** is rendered accurately by three models yet is conspicuously absent from Kandinsky’s generations. Together, these two cases concretely instantiate the dichotomy established in Sec. 5.2 in which cross-model agreement was found to be high for some concepts and limited for others.

Conversely, Fig. 10 presents a case of exaggeration, in which the concept **shadow under animal** is over-emphasized across generations. Although each shadow is individually plausible, their uniformly pronounced rendering across models—set against their subtler and more variable appearance in natural images—betrays an overactive prior. Notably, a deliberate search for concepts exaggerated by only a single model yielded no clear examples, indicating that, unlike suppression, exaggeration is approximately universal across the evaluated models.

Collectively, these examples make concrete how conceptual blindspots surface in generated outputs and confirm that our energy-based diagnostic reliably resolves both shared and model-specific failure modes. In particular, it exposes otherwise surprising limitations—such as the consistent, cross-model failure to reproduce clear or solid background elements like whitespaces on documents—and thereby demonstrates that certain failure patterns originate in architectural constraints or training-data biases that transcend the idiosyncrasies of any single model.

Informative as these aggregate analyses are, they raise a sharper question: do blindspots arise only in aggregate over many samples, or do they persist at the level of individual datapoints? Addressing



Figure 10: **Example of an exaggerated conceptual blindspot.** Four synthesized images $\mathcal{D}'_{\mathcal{X}}$, archetypal for the concept shadow under animal, shown alongside the corresponding natural image $\mathcal{D}_{\mathcal{X}}$ and the caption t used to generate them. The shadows beneath the animals are accen-

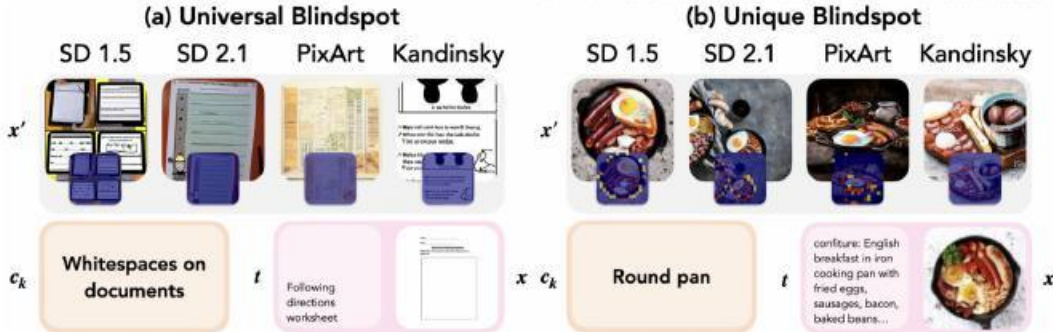


Figure 9: **Examples of suppressed conceptual blindspots.** Each natural image x , its representative concept c_k , and its caption t are shown alongside the four synthesized images x' produced by S_θ . The universal blindspot (solid white on documents) manifests in all four evaluated models, whereas the model-specific blindspot (pan) manifests in Kandinsky alone.

388 this question probes the mecha-
 389 nisms behind blindspots directly, dis-
 390 tinguishing prompt misinterpreta-
 391 tion from latent memorization and
 392 identifying where the two coin-
 393 cide.

395 5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

396 To move beyond population-level statistics, we examine the individual natural-
 397 generated image pairs whose $\delta(\cdot)$ values, averaged over all concepts, are largest
 398 and smallest—the two extremes at which the diagnostic is most informative.
 399 This datapoint-level view cleanly separates
 400 model successes from failures, the latter of
 401 which we find to arise predominantly from either prompt ambiguity or memorization artifacts.
 402 Fig. 11a shows pairs with near-zero average $\delta(\cdot)$,
 403 in which the generated image is conceptually
 404 indistinguishable from the original. Qualita-
 405 tive inspection reveals that this agreement stems
 406 not from faithful capture of the prompt semantics
 407 but from verbatim replication of memorized
 408 templates: the recurrence of near-identical vi-
 409 sual structures (e.g., clothing outlines or object
 410 arrangements) indicates that the model is re-
 411 producing over-frequent patterns lifted directly
 412 from the training data. Fig. 11b, conversely,
 413 shows pairs among the largest $\delta(\cdot)$, which signal
 414 substantial conceptual divergence between the
 415 synthesized and natural images. Some of these
 416 discrepancies trace to underspecified or noisy
 417 captions, but others expose genuine blindspots:
 418 the prompt describes a clear concept that is faith-
 419 fully present in $\mathcal{D}_X$, yet the model fails to realize
 420 it in $\mathcal{D}'_X$. Such cases demonstrate that even when language grounding is fully adequate, certain
 422 concepts lie beyond the model’s generative reach.



Figure 11: • **Datapoint-level conceptual alignment.** (a) Pairs with minimal energy difference, in which the model reproduces memorized training patterns. (b) Pairs with large energy difference, in which pronounced conceptual divergence arises either from prompt ambiguity or from genuine model limitations.

5.5 Analyzing Post-Training Effects

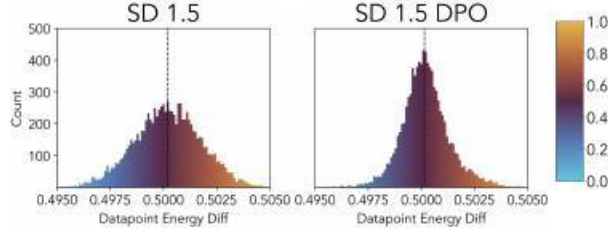


Figure 12: **Effect of DPO on concept fidelity.** Histogram of datapoint-wise energy differences between the synthesized and natural distributions for SD 1.5 with and without DPO. The DPO variant shifts mass toward smaller differences, providing direct evidence of improved conceptual alignment with the natural distribution.

Post-training protocols such as safety fine-tuning have been argued to reduce the diversity of model generations [140]. Because our pipeline isolates precisely the differences between a model’s generations and the ground-truth DGP, it provides a natural and direct instrument for quantifying the effect of DPO—a widely used safety fine-tuning protocol [141]. We compare two variants of SD 1.5, one trained with DPO and one without. For each image pair $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}'_{\mathcal{X}})$ we compute the ℓ_2 distance between their concept energy vectors, $\|\xi(\mathcal{D}'_{\mathcal{X}}) - \xi(\mathcal{D}_{\mathcal{X}})\|_2$. Fig. 12 presents the resulting histogram of datapoint-wise energy differences. The DPO variant attains both a lower median and a narrower spread, demonstrating that its generated concepts align more consistently with the ground-truth DGP. DPO thus acts to regularize the model’s concept distribution, steering its outputs toward the semantic content of the observed inputs. While our analysis does not disentangle the specific inductive biases DPO introduces, it furnishes direct empirical evidence that an objective favoring human-preferred generations also, and as a by-product, improves agreement with the training distribution. Crucially, this improvement is bidirectional: DPO attenuates *both* the over- and the under-activation of individual concepts relative to the natural baseline. These findings underscore the value of our pipeline for characterizing post-training interventions—not merely through output quality, but through the reshaping of the conceptual geometry of the model’s output space.

5.6 Conceptual Misalignment as a Function of Empirical Frequency

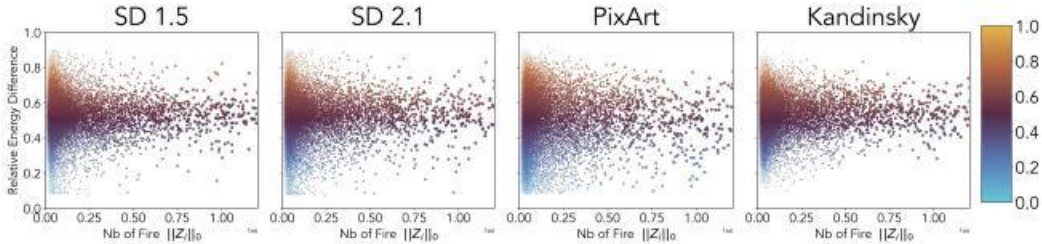


Figure 13: **Concept fidelity across the frequency spectrum.** Scatter plots relating concept frequency (x-axis) to energy difference (y-axis) for the four evaluated models. Each point is a concept, with its size proportional to its activation frequency; the trend, consistent across all four models, confirms that low-frequency, long-tail concepts concentrate the largest fidelity errors.

We hypothesized above that conceptual blindspots are not solely architectural artifacts but may also arise directly from the distributional properties of individual features. Here we test this hypothesis empirically, asking whether concepts that are rarely activated in natural images—those in the long tail of the data distribution—systematically induce blindspots in generative models. We process the natural dataset $\mathcal{D}_{\mathcal{X}}$ through the trained SAE and, for each concept k , compute its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$, the count of non-zero activations of concept i across all images. We then relate this frequency to the absolute energy difference observed in the generated outputs. Fig. 13 visualizes this relationship for all four models. A clear and consistent trend emerges: high-frequency concepts exhibit small energy discrepancies, whereas rare concepts—particularly suppressed ones ($\delta(k) < 0.5$)—incur substantial alignment errors. This establishes that many blindspots are not random model quirks but

451 the systematic downstream consequence of long-tailed concept distributions, and it indicates that
452 mitigating them will require not architectural changes alone but also data-level interventions such as
453 reweighting or augmentation.

454 6 Discussion

455 Our analysis uncovers numerous conceptual blindspots across four widely used generative image
456 models, and it does so through a single, unified framework. These results, moreover, only begin
457 to exploit the framework’s reach: each individual finding could sustain a dedicated investigation
458 of its own. Rather than pursue any single thread in depth, we have deliberately demonstrated the
459 breadth of our method and exploratory tool, which together support—out of the box—the systematic
460 identification of concepts a model struggles to generate, the detection of memorization artifacts, the
461 discovery of datapoints with insufficient captions, the quantification of post-training effects, and the
462 characterization of conceptual shifts across model architectures. This breadth positions the framework
463 as a general foundation on which follow-up work can build, whether by deepening the analysis,
464 widening the set of evaluated architectures, or investigating the root causes of conceptual blindspots.
465 A particularly promising direction is to endow the RA-SAE with hierarchical concept representations,
466 which would enable a correspondingly more nuanced, compositional analysis.

467 **Limitations.** Several limitations delimit the scope of our conclusions and, in doing so, clarify pre-
468 cisely what the framework does and does not establish. Because concept extraction and representation
469 rest on DINOv2 and the RA-SAE, our analysis is confined to the concepts these models encode:
470 concepts they represent poorly will necessarily escape detection. Furthermore, although a sample
471 of 10,000 images is substantial—and, as our concentration analysis rigorously shows, sufficient for
472 tight estimation of δ —it may not exhaust the long tail of rare concepts, nor fully resolve concept
473 co-occurrence and other higher-order compositional statistics.

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Appendix

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A Experimental Setup

This section specifies, in full, the experimental setup underlying our analysis of four widely used generative image models—SD 1.5/2.1, Kandinsky, and PixArt—all trained on LAION-5B or on its subsets and derivatives. The complete code is available at a software repository.

A.1 Observation Space

We construct the observation space by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which serves as our domain of natural images. Owing to concerns over CSAM and other unsafe content, the original release is no longer available; we therefore use a substitute release of a subset of the dataset with additional filtering of unsafe content, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>

The sampling procedure comprises three sequential steps: (1) loading the full LAION dataset through the Hugging Face `datasets` library; (2) validating each candidate by confirming a well-formed URL and resource availability via an HTTP HEAD request; and (3) sampling with replacement until 10,000 valid samples are obtained. This procedure yields D_G as a set of (x, t) tuples and guarantees an observation space of accessible image-text pairs—a non-trivial requirement, given that a large fraction of the original image URLs has become unavailable since release. Additional examples of synthesized images are provided in Appendix H

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} in exact one-to-one correspondence with D_G , producing triplets (x, x', t) . Concretely, for each of the 10,000 image-text pairs (x, t) in D_G , we condition the generative model g_θ on the caption t to synthesize the counterpart image x' .

Synthesis follows the standard text-to-image generation pipeline of each architecture, implemented with the Hugging Face `diffusers` library and executed at mixed precision (fp16). Unless stated otherwise below, all synthetic images are generated at 512×512 resolution with default parameters.

Stable Diffusion 1.5. The checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (which is a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>) is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5, employed in the analysis of Sec. 5.5 is identical to the baseline SD 1.5 implementation in every respect except that its UNet is replaced by the DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>

Stable Diffusion 2.1. The checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Kandinsky. The checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1> is used. Inference is performed using 100 inference steps, with the guidance scale fixed at 4.0.

PixArt. The checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1 For each evaluated model, we compute the energy difference $\delta(\cdot)$ across all 32,000 concepts, applying the sigmoid transformation with temperature $T = 0.8$ during normalization. The resulting values are rendered as log-scale density histograms with 100 bins spanning $[0, 1]$.

947 **Section 5.2** We embed the complete set of 32,000 concepts into two dimensions via UMAP applied
 948 to the sparse concept codes, coloring each point by its energy difference $\delta(\cdot)$ so as to foreground
 949 both suppressed and exaggerated blindspots. To quantify cross-model consistency, we then compute
 950 pairwise Pearson correlation coefficients between the $\delta(\cdot)$ vectors of every model pair, presenting the
 951 results as both scatter plots and correlation matrices. This analysis reveals whether blindspots cluster
 952 in conceptual space and cleanly disambiguates model-specific from universal blindspot patterns.

953 **Section 5.3** We rank all 32,000 concepts by their energy difference $\delta(\cdot)$ and manually examine the
 954 extrema at both ends of the ranking. For suppressed blindspots we select concepts with $\delta(\cdot) < 0.1$;
 955 for exaggerated blindspots, those with $\delta(\cdot) < 0.9$. The presented examples are annotated with textual
 956 concept descriptions obtained by inspecting each concept’s most activating images together with its
 957 spatial attention patterns. Our ongoing efforts to automate this concept interpretation are outlined in
 958 Appendix D

959 **Section 5.5** We compare SD 1.5 with and without DPO as follows: for each image pair (x, x') ,
 960 we compute the L2 norm of the difference between their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$,
 961 applying a sigmoid transformation with temperature $T = 0.8$ to the element-wise differences
 962 before averaging. This yields datapoint-wise energy differences that quantify precisely how far each
 963 generated image departs from its natural counterpart in concept space. The differences are then
 964 visualized as overlapping histograms that directly contrast the two model variants.

965 **Section 5.6** For each concept c_k , we count its empirical frequency $\|Z_{:,i}\|_0$ —the number of non-
 966 zero activations across the natural dataset—and apply a sigmoid normalization with temperature
 967 $T = 0.4$ to the energy differences $\delta(\cdot)$. The relationship is displayed as a scatter plot with empirical
 968 concept frequency on the x-axis and the sigmoid-transformed energy difference on the y-axis; point
 969 size is proportional to activation frequency, and point color encodes the magnitude of the energy
 970 difference.

971 A.4 • Datapoint Level Analysis

972 **Section 5.4** For each image pair (x, x') , we compute the L2 norm of the difference between
 973 their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$, yielding a single scalar measure of conceptual
 974 divergence per pair. Ranking the samples by this divergence exposes the two failure regimes at its
 975 extremes: minimal divergence flags potential memorization artifacts, whereas maximal divergence
 976 flags substantial conceptual failures. The ranking thereby enables targeted qualitative inspection of
 977 each specific failure mode.

978 A.5 • Co-occurrence Analysis

979 **Appendix F** For both the natural and the synthesized data, D_G and D_{g_0} , we analyze concept
 980 co-occurrence through the co-activation matrix $Z^T Z$, which encodes the pairwise correlations in
 981 concept usage. We then extract the dominant conceptual directions by eigendecomposition and
 982 assess the alignment between the natural and synthetic co-occurrence structures via cosine-similarity
 983 heatmaps between the top-100 eigenvectors of the two matrices. Each such 100×100 similarity
 984 matrix is displayed as a square heatmap: a perfect diagonal structure would signify identical principal
 985 concept axes, whereas off-diagonal mass reveals rotations and mismatches in the compositional
 986 geometry.

987 B Custom RA-SAE Training Details

988 This section documents, in full, the training details of our custom archetypal SAE (RA-SAE) [56],
989 open-sourced at a software repository.

990 **Dataset.** The autoencoder is trained on the complete ImageNet-1k training split (≈ 1.28 M RGB
991 images). Each image is encoded into 261 visual tokens by DINOv2 [135], and these tokens are
992 fed to the SAE without class or position embeddings. The training corpus therefore comprises
993 $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$ tokens.

994 **Dictionary.** The dictionary spans 32,000 concept dimensions. Sparsity is enforced by top- k masking
995 with $k = 5$: for each input, all activations outside the five largest are set to 0. Weights are initialized
996 via Xavier/Glorot, and training is conducted at mixed precision (fp16), with the final ten epochs
997 performed at full precision to sharpen convergence.

998 **Optimizer and schedule.** The model is trained for 50 epochs with the AdamW optimizer ($\beta_1 = 0.9$,
999 $\beta_2 = 0.999$) and weight decay 10^{-5} . The learning rate follows a linear warm-up over the first 5%
1000 of steps and a subsequent cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. The objective is an
1001 MSE reconstruction loss augmented by an auxiliary term, weighted by $\lambda = 10^{-5}$, that penalizes
1002 activations which never enter the top- k set.

1003 C Computational Resources

1004 This section itemizes the GPU resources consumed by the training and experiments reported in this
1005 paper, which total approximately 202 GPU-hours on NVIDIA H100 and H200 hardware.

1006 **RA-SAE.** Training required approximately 24 GPU-hours on three NVIDIA H100s.

1007 **Synthesized images.** Generating the full $D_{g\theta}$ (see Appendix A) took roughly 5 hours per generator
1008 when distributed across four NVIDIA H200 GPUs; aggregated across five generators, this amounted
1009 to approximately 100 GPU-hours on the H200.

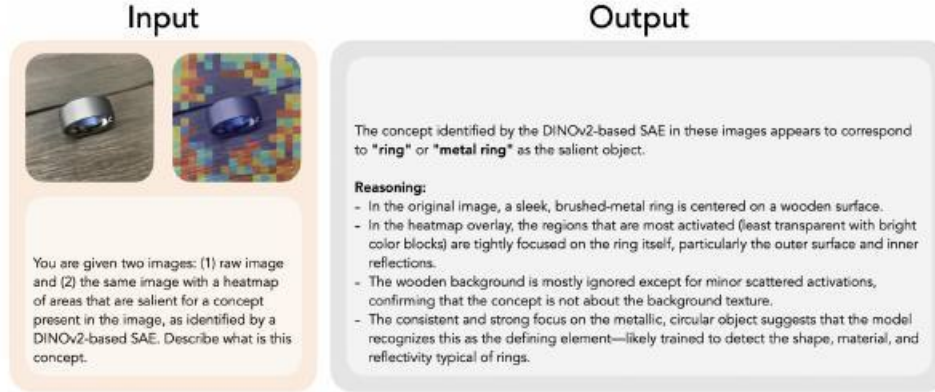
1010 **• Distribution-level analysis.** Extracting distribution-level energy differences took about 3 hours
1011 per generator on a single NVIDIA H200 GPU (~ 15 GPU-hours in total).

1012 **• Datapoint-level analysis.** Computing datapoint-level energy differences, ranking concepts per
1013 datapoint, and ranking datapoints per concept together took approximately 3 hours per generator on
1014 one H200 GPU (~ 15 GPU-hours in total).

1015 D Autointerpretability

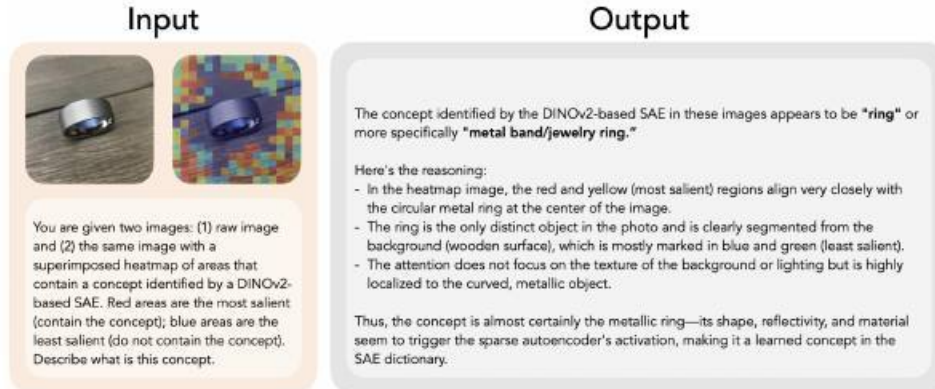
1016 Toward automating the interpretation of the concepts identified by the RA-SAE, we conducted a
 1017 qualitative evaluation of the ability of Vision-Language Models (VLMs) to describe them. Specifically,
 1018 we queried ChatGPT 4o through the OpenAI API using a range of prompts and image formats, each
 1019 derived from an image x'' in which the target concept was salient, and recorded the VLM’s description.
 1020 This section reports the findings of that exploration. For evaluation we permitted the VLM to emit
 1021 its full reasoning; a scalable deployment would suppress this reasoning through a minimal prompt
 1022 modification requesting the description alone.

1023 We first supplied the VLM with two images: a raw version of x'' and a version with a superimposed
 1024 heatmap localizing the intensity of the concept:



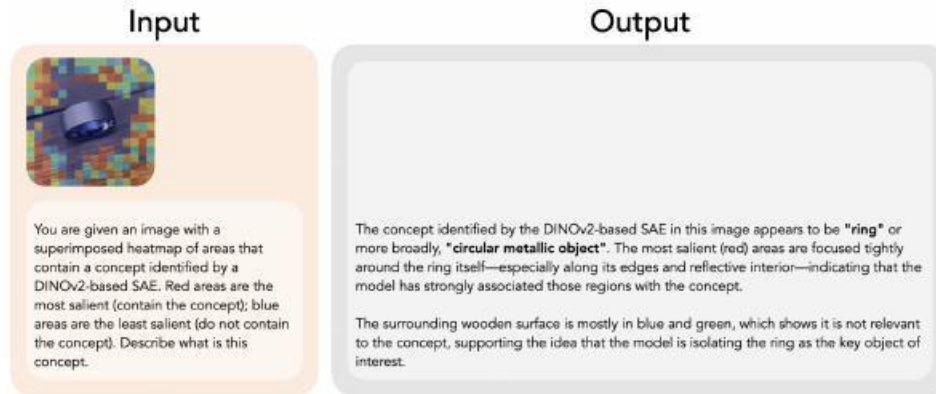
1025 As the example shows, the VLM failed to interpret the heatmap correctly: rather than describing the
 1026 wood texture in the background—the genuinely salient region—it described the foreground ring, to
 1027 which the RA-SAE assigned no activation whatsoever.

1028 To ascertain whether this failure was merely an artifact of prompting, we repeated the experiment
 1029 with an expanded prompt that explained the heatmap in greater detail:



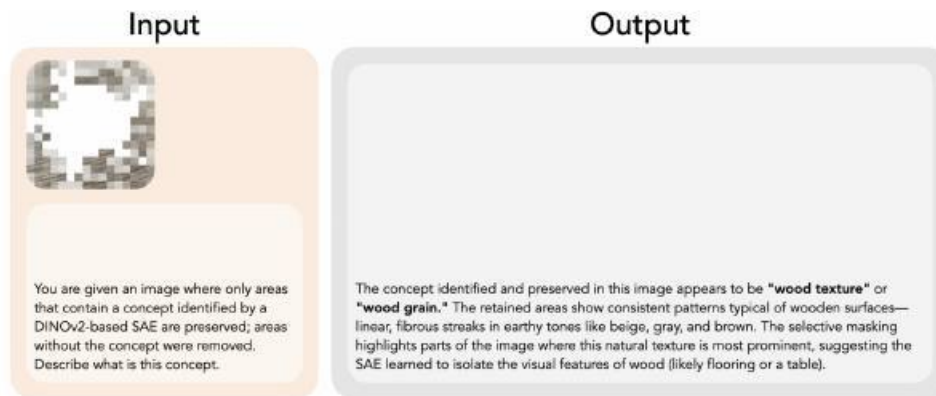
1030 This modification did not resolve the issue: the VLM again identified the ring as the prominent
 1031 concept. Several further prompting strategies—including ones that described the heatmap in still
 1032 greater depth, for instance by supplying the code used to generate it—produced comparable results.

1033 We next provided the VLM with the heatmap-superimposed x'' alone, accompanied by a similar
1034 prompt explaining how to interpret the heatmap:



1035 Once again the VLM failed to identify the wooden background and instead focused on the ring;
1036 varying the prompting strategy for this image format left the results essentially unchanged.

1037 These observations lead us to conclude that the heatmap alone is insufficient to direct the VLM's
1038 attention to the salient region, since the presence of less relevant regions distracts the model. To
1039 remove this distraction, we replaced the heatmap-superimposed x'' with an alpha-masked version of
1040 x'' :



1041 As the example shows, the VLM now correctly identified wood texture as the dominant concept,
1042 and we observed consistent success across a broad range of concepts and images.

1043 The formatting of x'' and the prompt shown above yielded the most reliable results in our qualitative
1044 evaluation. We stress, however, that this evaluation is bounded by its qualitative nature—owing to the
1045 absence of ground-truth annotations—and by its focus on a single VLM. We anticipate that future
1046 work on the autointerpretability of SAE concepts will build directly on and extend this analysis.

1047 E Additional Results: Model-Specific Blindspots



Figure 14: Examples of conceptual blindspots in **Stable Diffusion 1.5**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by its highest absolute activation, with the concept’s spatial heatmap superimposed to localize where the concept is expressed.

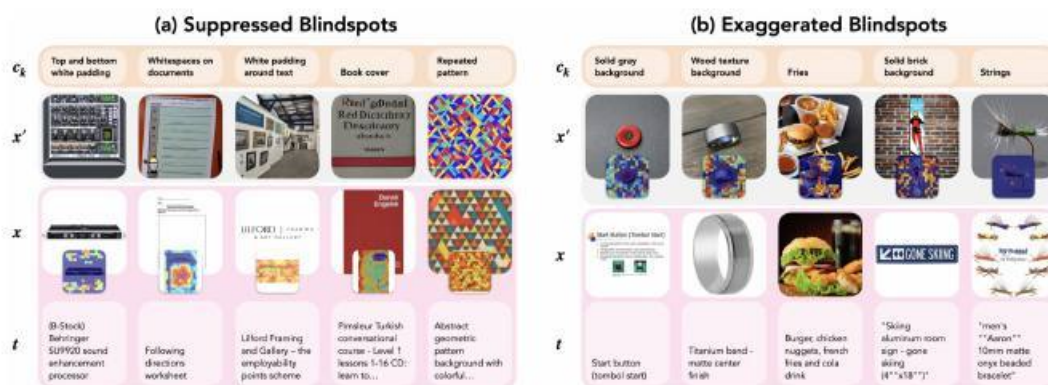


Figure 15: Examples of conceptual blindspots in **Stable Diffusion 2.1**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by its highest absolute activation, with the concept’s spatial heatmap superimposed to localize where the concept is expressed.

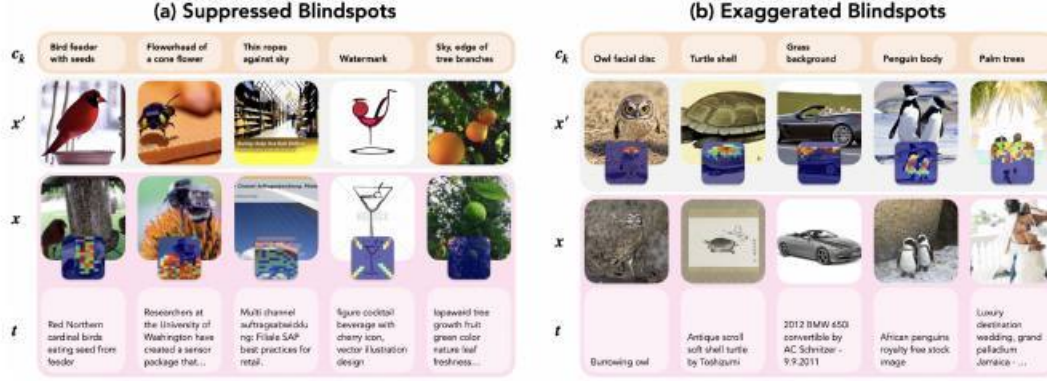


Figure 16: Examples of conceptual blindspots in **Kandinsky**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by its highest absolute activation, with the concept’s spatial heatmap superimposed to localize where the concept is expressed.



Figure 17: Examples of conceptual blindspots in **PixArt**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by its highest absolute activation, with the concept’s spatial heatmap superimposed to localize where the concept is expressed.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Our analysis has so far centered on individual concept activations. Visual scenes, however, are rarely composed of isolated concepts; they are organized by rich co-occurrence patterns that encode compositional semantics. We now examine whether, and to what extent, generative models capture this higher-order structure by analyzing the co-activation matrix $Z^T Z$, whose entries are the pairwise correlations in concept usage.

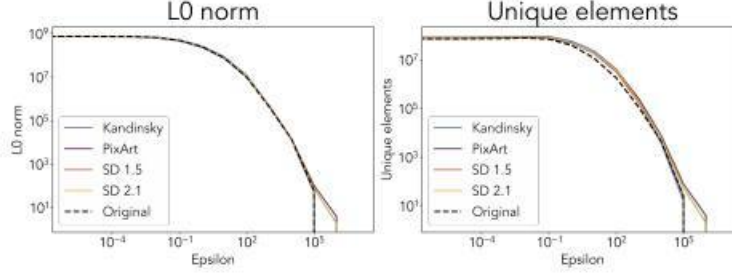


Figure 18: **Sparsity and structural divergence.** Left: the ℓ_0 norm of the co-occurrence matrix ZZ^T as a function of the threshold ϵ , indicating how many entries remain active in each model. Right: the number of unique entries in the synthesized distribution relative to the natural distribution. Every evaluated model faithfully preserves the global sparsity structure yet diverges in its specific activation content.

Notably, at the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As Fig. 18 (left) shows, the ℓ_0 norm of $Z^T Z$ —thresholded across a range of ϵ values—tracks closely between the natural and synthesized distributions for every model. The gross connectivity of the conceptual graph—that is, which concepts co-activate at all—is thus well preserved. Formally, $Z^T Z$ may be read as the adjacency matrix of a weighted, undirected graph over concepts whose edge weights encode co-activation strength across the dataset.

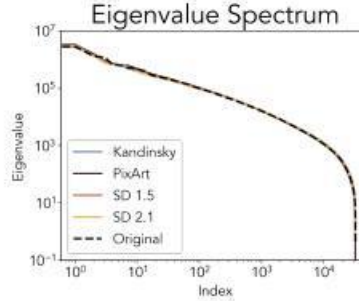


Figure 19: **Spectral structure of co-occurrence.** Log-log plot of the eigenvalue spectra of the co-occurrence matrices ZZ^T for each model and for the natural distribution. Every evaluated model faithfully reproduces the heavy-tailed decay of the natural distribution.

The specific content of these co-activations, however, diverges: as Fig. 18 (right) shows, a substantial fraction of the entries in the model-generated $Z^T Z$ are absent from the natural baseline. The capacity for compositionality is thus retained even as the identity of the active pairings shifts—a shift that plausibly reflects model-specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we turn to spectral analysis. Fig. 19 presents the eigenvalue spectra of the co-occurrence matrices for each model and for the natural distribution. Every spectrum exhibits heavy-tailed, power-law-consistent decay, establishing that generative models preserve both the overall rank structure and the allocation of variance across conceptual dimensions.

We further examine the alignment of the dominant conceptual directions via cosine-similarity heatmaps between the top 100 eigenvectors of the synthesized and natural co-occurrence matrices (Fig. 20). Although every model exhibits partial diagonal alignment—confirming a substantial

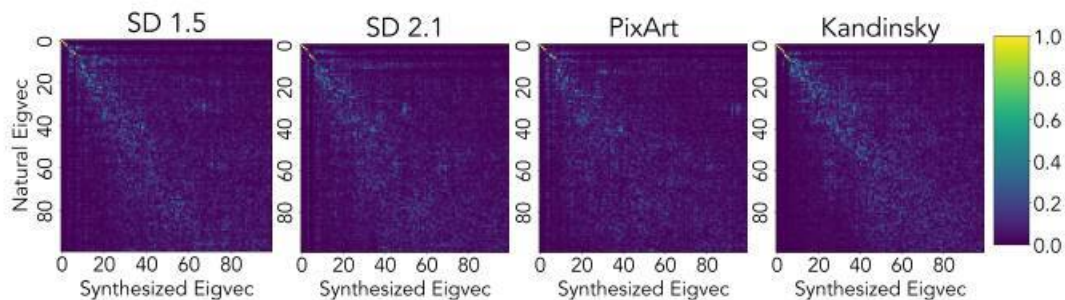


Figure 20: ● **Concept basis similarity.** Cosine-similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure indicates alignment of the dominant conceptual directions, whose degree varies systematically across the four models.

1073 overlap in the principal concept axes—the off-diagonal entries expose rotations and mismatches in
 1074 the higher modes, marking clear and structured deviations in compositional geometry.

1075 Together, these findings establish a nuanced but coherent picture: diffusion models approximate the
 1076 global shape of concept co-activation remarkably well, yet deviate in subtle and structured ways once
 1077 examined through the spectral lens. Such higher-order discrepancies furnish a plausible mechanism
 1078 for the observed failures in generating coherent multi-object scenes and relational concepts.

1079 G Additional Results: Blindspot Stress Testing

1080 To stress-test the blindspots identified by our method, we assembled a range of prompts describing
 1081 each blindspot, used them to generate many images, and contrasted the outputs of the models in
 1082 which the concept was flagged as a blindspot against those of the models in which it was not.

1083 Concretely, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1084 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1085 *text-to-image model. Return these as a list of strings in Python.* We generated five images per
 1086 prompt and analyzed them with our custom RA-SAE model (Appendix B), which ranked them by
 1087 the intensity with which the target concept appeared; all images were then manually reviewed to
 1088 determine whether the blindspot was successfully depicted.

1089 As the following examples of suppressed concepts illustrate, although fragments of the target concept
 1090 occasionally surfaced—for instance a holder or string for the `bird feeder` blindspot, or a round
 1091 hole for the `glossy DVD disc` blindspot—the models systematically failed to render the complete
 1092 concept. This outcome directly corroborates our method’s assessment and thereby validates the stress
 1093 test as an independent check on the diagnostic.

1094 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Images generated from a variety of prompts involving the `bird feeder` concept during stress testing. Our method flags this concept as a suppressed conceptual blindspot in Kandinsky, and the generations bear this out precisely: the model cannot render it. SD 1.5, for which the concept was not flagged as a blindspot, generates it successfully—exactly as the diagnostic predicts.



Figure 22: Images generated from a variety of prompts involving the `glossy DVD disc` concept during stress testing. Our method flags this concept as a suppressed conceptual blindspot in SD 1.5, and the generations bear this out precisely: the model cannot render it. Kandinsky, for which the concept was not flagged as a blindspot, generates it successfully—again matching the diagnostic’s prediction.

1096 **H Additional Examples: Synthesized Images**



Figure 23: Additional image-caption pairs from LAION-5B, each shown alongside the images synthesized from the identical prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1097 I Concentration bounds for δ

1098 In our experiments we estimate $\delta(k)$ from $n = 10,000$ paired samples per concept. This sample
 1099 budget is deliberately modest, which raises the natural question of whether it suffices for reliable
 1100 estimation. We answer it decisively in the affirmative by deriving a concentration bound on the
 1101 empirical estimator $\hat{\delta}_n(k)$ via McDiarmid’s inequality [142]. The bound is tight and establishes that
 1102 even a comparatively small sample yields fast, accurate estimates of concept bias.

1103 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1104 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1105 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1106 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1107 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1108 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1109 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1110 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1111 By McDiarmid’s inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1112 which proves the claim. □

1113 In practice, the concept scores $\delta(k)$ are sparse: the majority concentrate near zero and only a few
 1114 reach values as large as 10. The concentration bound guarantees that even at the largest observed
 1115 biases, a sample of $n = 10,000$ yields estimates $\hat{\delta}_n(k)$ that deviate from the true value by no more
 1116 than a small ε with high probability. This rigorously justifies our sampling strategy and confirms that
 1117 accurate bias measurements are fully attainable from limited data.

1118 J Monotonicity and Calibration-Free Interpretation of δ

1119 Our objective when analyzing blindspots is to rank concepts by the severity of their generative bias,
 1120 a purpose the score $\delta(k)$ serves directly in practice. One might reasonably ask whether this score
 1121 distorts the ranking relative to more elementary quantities such as the energy gap or the odds ratio.
 1122 The following result settles the question conclusively: $\delta(k)$ is a strictly increasing reparameterization
 1123 of both, and therefore inherits their ordering exactly, so that ranking concepts by $\delta(k)$ requires no
 1124 calibration whatsoever.

1125 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1126 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_{\mathcal{X}}}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{X}}}[\xi_k(\mathbf{x})],$$

1127 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1128 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1129 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1130 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1131 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1132 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1133 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1134 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1135 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1136 concepts.

1137

□

1138 Ranking concepts by $\delta(\cdot)$ is therefore strictly equivalent to ranking them by the energy gap or by the
 1139 conceptual generation odds ρ_k : no calibration is necessary, and all three quantities induce one and
 1140 the same total ordering over concepts.

K Stability of FID Under SAE Embeddings

In this section we establish an exact quantitative relationship between the Fréchet Inception Distance (FID) computed in the original activation space (of dimension d) and the FID computed after applying a possibly overcomplete SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout, we assume D has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

We begin by recalling an elementary fact: when D is orthogonal and square ($k = d$), it acts as an isometry between $\mathbf{A}$ and $\mathbf{Z}$, so that FID is preserved exactly. This regime is unrealistic, however, and we therefore turn to the overcomplete case, in which we bound the FID distortion by the extreme singular values of D . We work throughout with the Wasserstein-2 metric $\mathcal{W}_2$, exploiting the fact that FID is precisely $\mathcal{W}_2^2$ specialized to Gaussians.

For a probability measure μ on $\mathbb{R}^d$, we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) = \mu(D^{-1}z)$. Let $\sigma_{\min}$ and $\sigma_{\max}$ denote the minimal and maximal singular values of D —equivalently, the square roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

Empirically one typically finds $\sigma_{\min}, \sigma_{\max} \approx 1$, although the proof relies on no such assumption. We first record a simple lemma for the non-overcomplete case.

Lemma 1 (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

Proof. Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical. $\square$

This case is nonetheless unrealistic, since SAEs rely on overcompleteness precisely in order to extract meaningful, interpretable concepts. In the overcomplete regime D is no longer orthonormal, yet it can remain column-orthonormal; we leverage exactly this property to bound the distortion in terms of the extremal singular values of $D^\top D$.

Theorem 6 (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices $A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

Proof. Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have, by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1175 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1176 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1177 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1178 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in $(\mathbf{A}, \mathbf{A}')$ (resp. $(\mathbf{Z}, \mathbf{Z}')$)
 1179 finishes the proof. $\square$

1180 In essence, Theorem 6 establishes that applying a column-orthogonal overcomplete SAE dictionary
 1181 distorts the Fréchet Inception Distance by no more than the square of its extremal singular values.
 1182 When $\mathbf{D}$ is nearly orthogonal—and empirically we consistently found $\sigma_{\min}, \sigma_{\max} \approx 1$ —the result
 1183 guarantees that FID measured in the SAE feature space is, for all practical purposes, indistinguishable
 1184 from the canonical FID. $\square$

1185